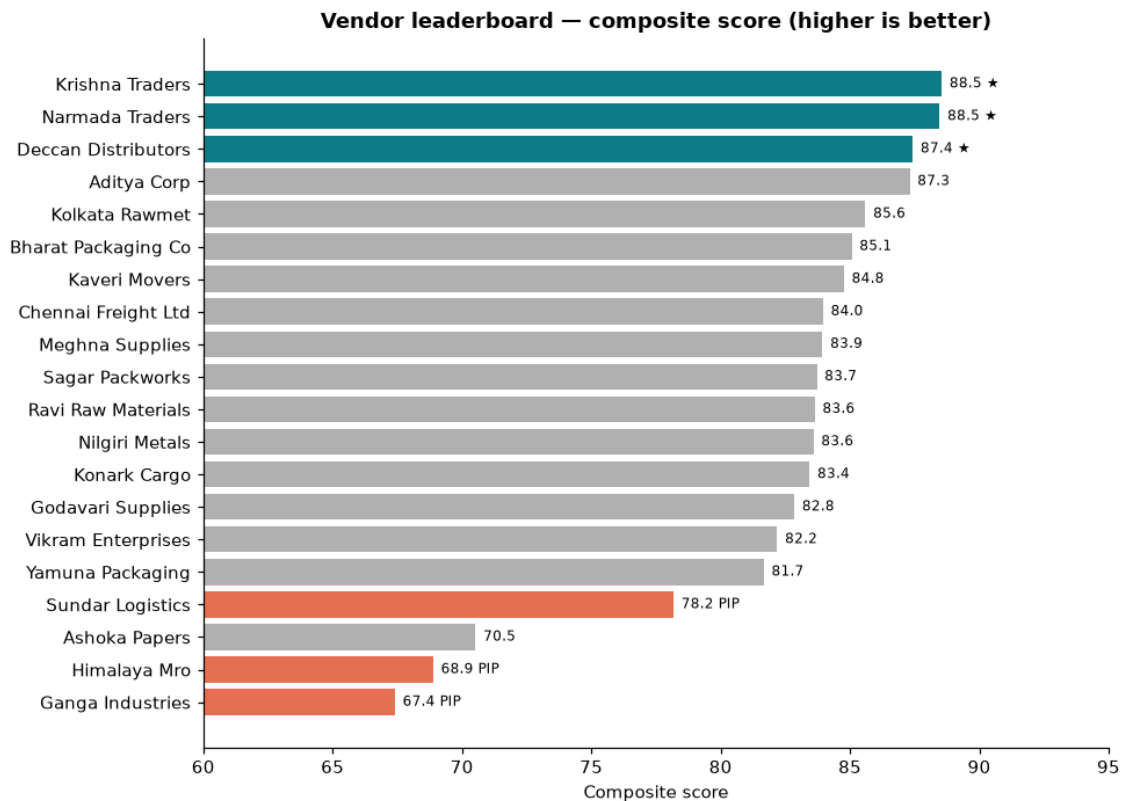


Vendor Performance Analytics

NorthBridge Supplies Pvt. Ltd. — Jan–Sep 2025

This report supports quarterly vendor-review decisions. It identifies candidates for a Performance Improvement Plan (PIP) and vendors eligible for continued or expanded business, based on delivery, quality, and cost performance across ~1,400 purchase orders.

Prepared: 09 Sep 2026 • Author: Data Analytics (fresher project) • Currency: INR



Bottom line: Three vendors are recommended for PIP review — **Ganga Industries, Himalaya Mro, Sundar Logistics**. Three vendors are recommended for continued/expanded business — **Krishna Traders, Narmada Traders, Deccan Distributors**.

1. Business problem

NorthBridge Supplies procures from 20 vendors across four categories (Packaging, Raw Materials, Logistics, MRO) and four regions (North, South, East, West India). Procurement leadership needs a quarterly, evidence-based view to answer:

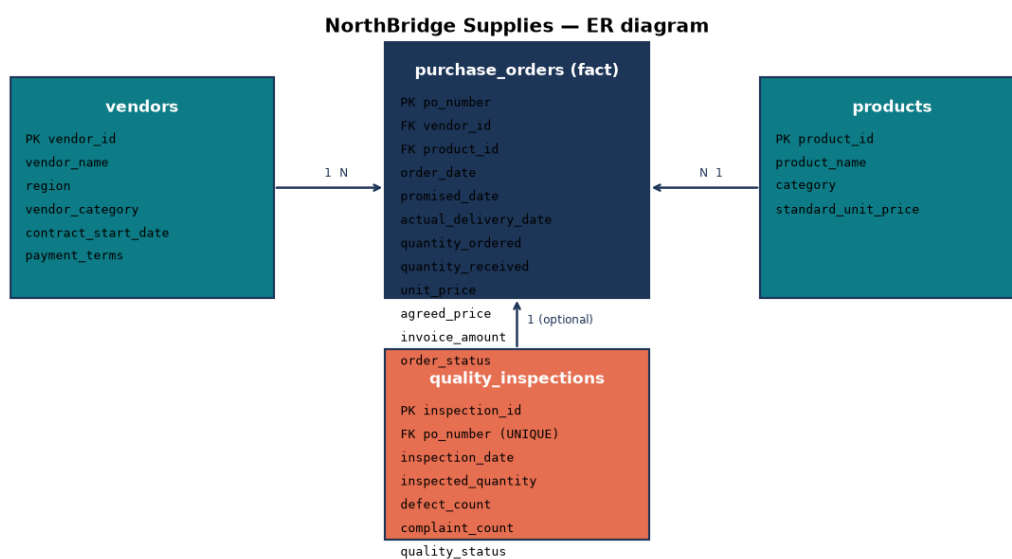
- Which three vendors should be placed on a Performance Improvement Plan (PIP) this quarter?
- Which vendors should be considered for continued or increased business?

The analysis **supports** the decision — it does not automatically recommend replacement. Sole-source vendors and total-cost-of-ownership considerations are outside the scope of this project.

2. Data & tools

Dataset: 1,400 purchase orders (2025-01-01 → 2025-09-28), 20 vendors, 16 products, total spend ■110.90 Cr. Quality inspections are performed on a subset of POs (~60% coverage), reflecting a realistic sampling-based QC process. Dataset is synthetic but deliberately dirty: duplicated POs, inconsistent vendor-name casing, ~1% missing delivery dates, ~2% missing defect counts, and one seasonal dip.

Stack: PostgreSQL 15 (16-compatible SQL), Python 3.11 (pandas, numpy, matplotlib, psycopg2, Faker), Power BI Desktop (dashboard mocked as static PNGs in this repo; DAX measures shipped as a separate file).



3. Methodology & KPI definitions

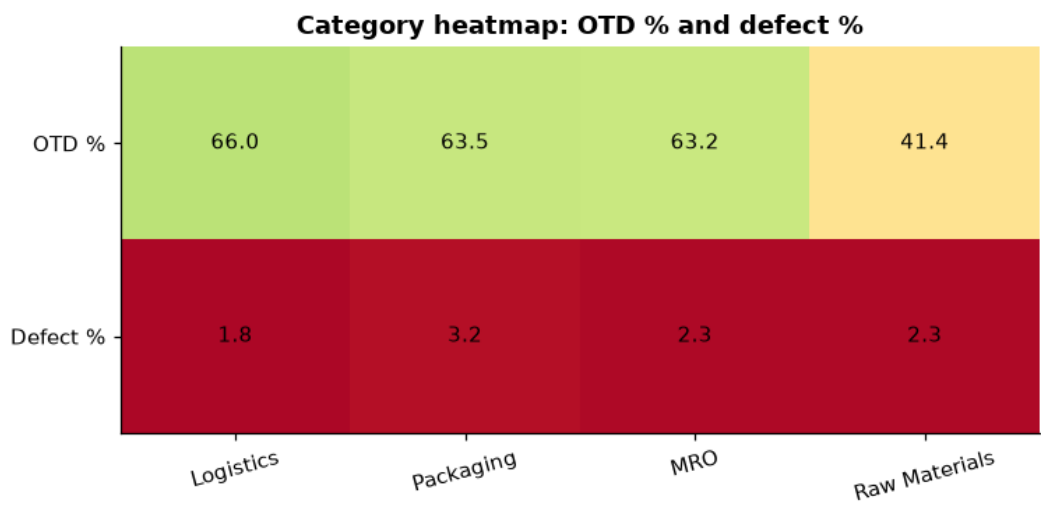
All KPIs are computed on cleaned tables (*_clean). The cleaning notebook logs every decision (duplicate PO drop, casing fix, zero-price flag, over-shipment flag, missing-defect handling).

KPI	Formula	Scope / notes
On-Time Delivery %	$\text{count}(\text{actual} \leq \text{promised}) / \text{count}(\text{delivered})$	Rows with missing actual date excluded from denominator
Defect Rate %	$\Sigma \text{defect_count} / \Sigma \text{inspected_qty} \times 100$	Inspected POs only. Missing defect filled 0 only when status=Partial
Price Variance %	$(\text{unit_price} - \text{agreed_price}) / \text{agreed_price} \times 100$	Rows with agreed_price=0 or NULL excluded
Lead Time (days)	$\text{actual_delivery_date} - \text{order_date}$	Missing actual date excluded
Fill Rate %	$\text{quantity_received} / \text{quantity_ordered} \times 100$ (capped at 100)	Over-shipments flagged separately in data-quality view
Composite Vendor Score	$0.40 \cdot \text{OTD} + 0.40 \cdot (100 - \text{Defect}\%) + 0.20 \cdot (100 - \text{FillRate}\%)$	Eligible Price Vars. Weights academic, not industry-standard

Disclaimer: The 40/40/20 composite weights were chosen for this academic project. They are not an industry-standard formula.

4. Findings

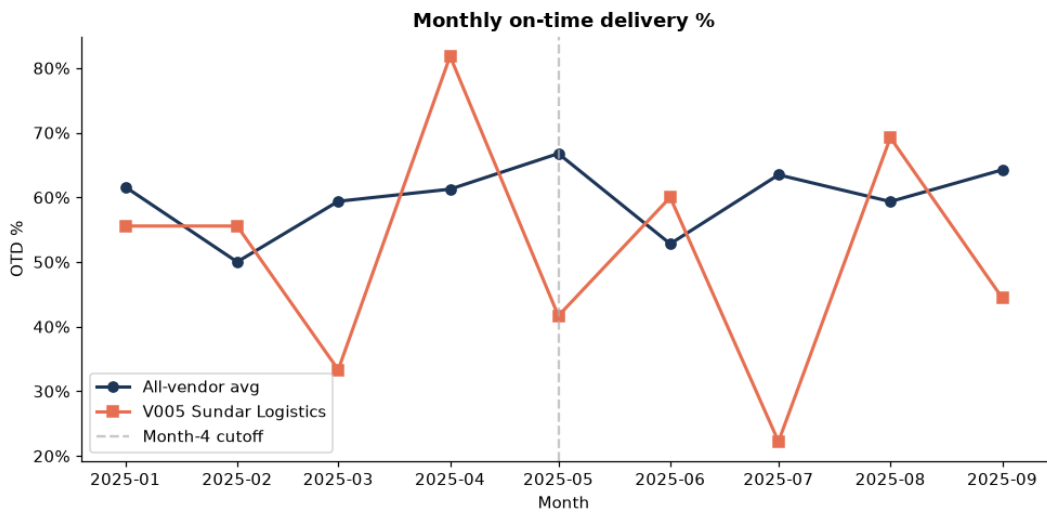
4.1 Overall picture. Across 1,400 POs, average vendor OTD is 58.7% and average defect rate is 2.47%. Category-level performance shows Raw Materials with the weakest OTD, driven by longer promised lead times AND two chronically-late vendors.



4.2 Delivery patterns. The delay histogram is roughly centred around zero but with a long right tail — a handful of POs are severely late. When a vendor misses, they tend to miss badly.



4.3 Quality trend — one vendor decline. Comparing pre-M4 and post-M4 windows, **Sundar Logistics's** defect rate rose from 1.94% to 10.32% (Δ +8.4 pts). Other vendors move within noise ranges.



Top 5 vendors by quality-decline delta:

Vendor	Name	Pre-M4 inspection	Post-M4 inspection	Pre %	Post %	Δ pts
V005	Sundar Logistics	26	31	1.94	10.32	8.38
V002	Ganga Industries	8	9	1.28	3.37	2.1
V019	Narmada Traders	4	10	0.0	1.78	1.78
V014	Ashoka Papers	4	8	0.38	2.14	1.75
V004	Meghna Supplies	34	43	1.62	2.68	1.06

5. Recommended action list

5.1 PIP candidates (bottom-3 eligible with a delivery or quality issue).

Vendor	Name	POs	OTD %	Defect %	Fill %	Score
V002	Ganga Industries	28	21.43	2.35	97.23	67.41
V010	Himalaya Mro	28	25.0	2.16	98.11	68.89
V005	Sundar Logistics	87	52.33	6.14	96.94	78.18

- **Ganga Industries** (V002, Raw Materials): OTD 21.4% (benchmark 60.3%), defect 2.35% (benchmark 2.53%), composite 67.4. Selection driver: OTD below benchmark.
- **Himalaya Mro** (V010, Raw Materials): OTD 25.0% (benchmark 60.3%), defect 2.16% (benchmark 2.53%), composite 68.9. Selection driver: OTD below benchmark.
- **Sundar Logistics** (V005, Packaging): OTD 52.3% (benchmark 60.3%), defect 6.14% (benchmark 2.53%), composite 78.2. Selection driver: OTD below benchmark and defect rate above benchmark.

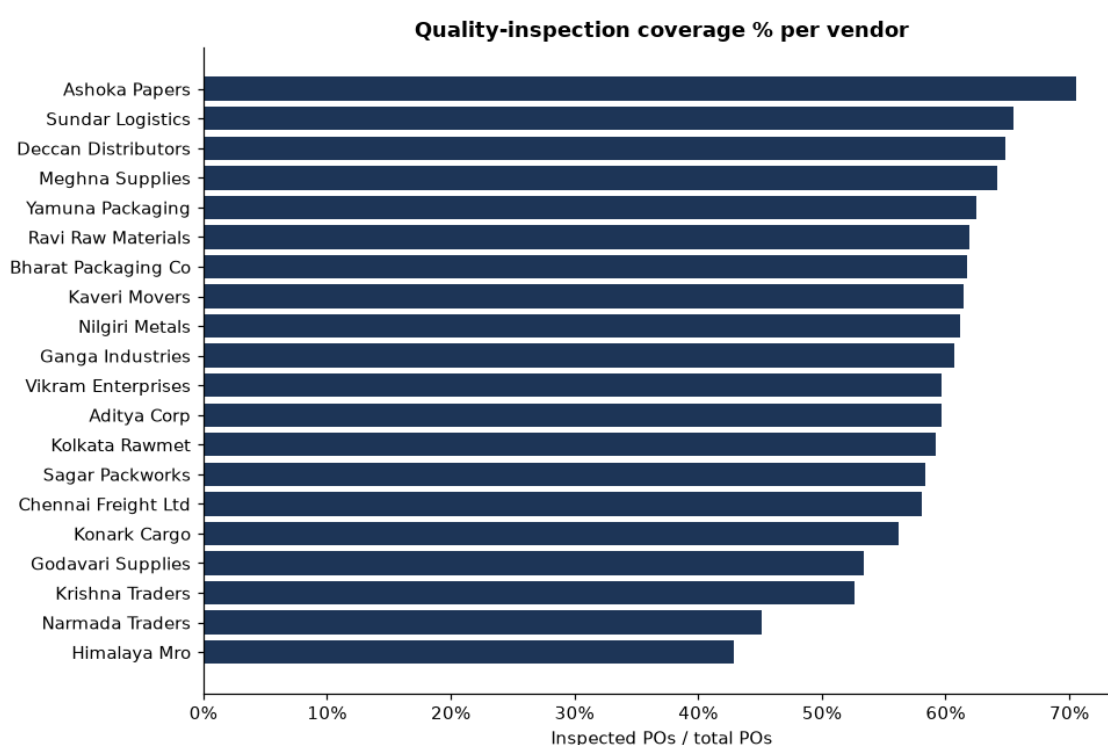
5.2 Reward candidates (top-3 eligible).

Vendor	Name	POs	OTD %	Defect %	Fill %	Score
V003	Krishna Traders	38	73.68	1.38	97.36	88.55
V019	Narmada Traders	31	74.19	1.59	97.18	88.47
V013	Deccan Distributors	71	71.43	1.63	96.83	87.43

- **Krishna Traders** (V003, Packaging): OTD 73.7%, defect 1.38%, fill 97.4%, composite 88.5. Recommended for continued business; consider volume expansion subject to sole-source-risk check outside this analysis.
- **Narmada Traders** (V019, Packaging): OTD 74.2%, defect 1.59%, fill 97.2%, composite 88.5. Recommended for continued business; consider volume expansion subject to sole-source-risk check outside this analysis.
- **Deccan Distributors** (V013, Logistics): OTD 71.4%, defect 1.63%, fill 96.8%, composite 87.4. Recommended for continued business; consider volume expansion subject to sole-source-risk check outside this analysis.

6. Limitations

- No sole-source vendor logic — a low-scoring vendor may still be the only supplier for a critical SKU. Any PIP decision must be validated against category-manager input.
- Total-cost-of-ownership is not modelled. Price variance is an estimate; it ignores freight, quality-reject rework, and financing effects.
- Quality metrics reflect only inspected POs. Vendors with low inspection coverage (see chart below) have less reliable quality scores.
- The dataset is synthetic. Patterns are illustrative and not real market behaviour.
- Composite-score weights (40/40/20) are academic choices; an industry-standard scorecard would use category-specific weights, spend-weighting, and sensitivity analysis.
- Small-sample vendors (< 20 POs) are excluded from the leaderboard eligibility. This is why *Ashoka Papers* (17 POs) appears with a low score but is not shortlisted.



7. Appendix — chart index

Figures generated: order volume by vendor, monthly OTD trend, defect distribution, delivery delay histogram, price variance by category, vendor leaderboard, category heatmap, delivery buckets, inspection coverage, ER diagram, and two Power BI dashboard mockups (Executive Summary, Vendor Deep-Dive).